Position, distance, displacement

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish a location, a total path length, and a change in position.

SUCCESS CRITERIA - I can...

1. Label the origin, start, turn and finish.
2. Calculate path distance and signed displacement separately.
3. Explain why position and displacement can differ.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A cart is at 3, 8, 13 m at 0, 1, 2 s. Predict its position at 7 s. State an assumption and a measurement that could challenge it.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta x = x_f - x_i$; distance = sum of path lengths. Choose an origin and positive direction. i = initial; f = final.

Worked reasoning: A walker starts at 2 m, reaches 6 m, then ends at 4 m. Distance=4+2=6 m. Displacement=4-2=2 m right.

Your turn: The walker goes from the 2 m floor mark to the 6 m mark, then back to 4 m. On the number line, draw the two legs above the axis and a separate start-to-finish arrow below it. Label each length.

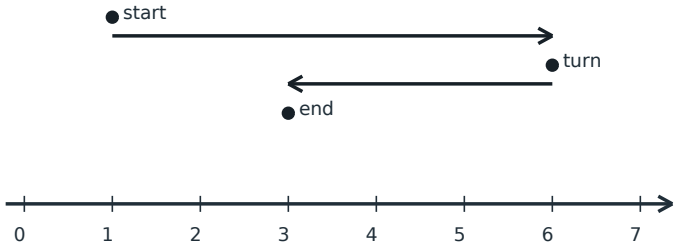


Position (m); right is positive

Marzano check - circle: ① ② ③

03 A walk along the hallway

You walk along a hallway marked in meters. Start at 1 m, turn around at 6 m, and stop at 3 m. Label the start, turn and finish on the route.



Position (m); right is positive

Marzano check - circle: ① ② ③

04 Count the whole path

For the route in 3, find each leg and the total distance. First leg: ____ m. Return leg: ____ m. Total: ____ m.

Marzano check - circle: ① ② ③

05 Connect start to finish

For the same route, draw one arrow from start to end. Give the displacement and final position. Are they the same number?

Support: I end ____ m to the ____ of where I started.

Marzano check - circle: ① ② ③

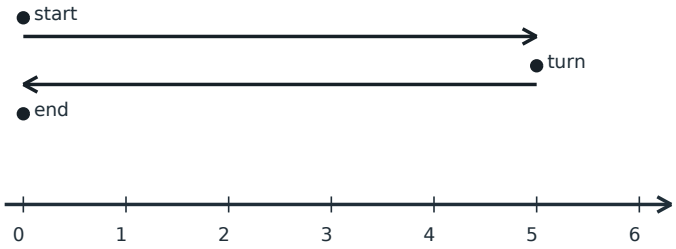
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Continue down each column.

06 A complete trip back

You leave a doorway, walk 5 m east, then return along the same path. Find distance and displacement. Use the route picture to explain why only one is zero.



Position (m); right is positive

Marzano check - circle: 1 2 3

07 Fix the wording

A student says, “My displacement was 8 m because I walked 8 m altogether.” Must that be true? Use a short route or sketch to explain.

Marzano check - circle: 1 2 3

08 Review: connect an earlier idea

A cart is observed at 3, 8, 13 m, one second apart. Record two patterns and point to values that support each.

Time (s)	Position (m)
0	3
1	8
2	13

Marzano check - circle: 1 2 3

09 Exit check

A robot starts at the 2 m mark, travels to 7 m, then returns to 5 m. Sketch the route. Find total distance, displacement with direction, and final position.



Position (m); right is positive

Marzano check - circle: 1 2 3

10 STRETCH: test the model

Design two routes that start at 1 m and end at 4 m: one of distance 3 m and one of distance 7 m. Draw and check both.



Position (m); right is positive

Marzano check - circle: 1 2 3